

Regime Viability on the Northern Cod Stock: An Applied Certification of the Obstruction Programme's Machinery on Locked Public Data

The programme's applied flagship (P5)

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Abstract

The obstruction programme's limitation L4 states that no programme result is calibrated against real stocks or sensor data. This paper closes that limitation with an exact certification of the programme's viability and obstruction machinery on the Northern cod stock (NAFO 2J3KL), from provenance-locked public data (DFO 2016 Table A2; the xteNCAM 2025 reassessment Table 17; the Regular et al. 2025 landings table; the capelin acoustic series), every number regenerated by the edition's script in exact rational arithmetic. The certified findings are stark. The 1983–1989 “healthy” decade's worst realized net multiplier is $\frac{41800}{45147} < 1$: no removals policy certifies indefinite above-floor viability even before the collapse — the reference decade was already not robustly viable at the 276 kt auxiliary threshold, which is what the LRP's construction (the decade's own mean, certified to $\pm 6.4\%$) encodes. Across 1992–1993, with realized removals falling to 11,392 t, the stock's realized net multiplier was $\frac{2021}{7639} \approx 0.265$: the decline proceeded with the fishery effectively closed, a witnessed obstruction in the dynamics, not the catch level. The breach-time certificates match the realized timeline exactly (below the 276 kt threshold within two years of the 1991 reading, as realized in 1993), and the capelin covariate collapses by a certified factor of 138/5783 in 1991 — one step before the mortality regime crosses $M = 1$ — the probe datum that makes monitoring, not catch, the viable instrument. Every claim is a certificate about the realized windows, not a forecast.

Keywords: viability; obstruction; regime shift; Northern cod; exact computation; monitoring design; limit reference points

0.1 Introduction and charter

The programme's applications note records, as limitation L4, that “no result in the programme is calibrated against real schedules, stocks, or sensor data” and that applied calibration is an open owner-side item. This paper is the calibration. It applies the programme's viability and obstruction machinery — survivable sets, robust multipliers, breach-time certificates, and probe design — to the

Northern cod stock (NAFO divisions 2J3KL), using exclusively the provenance-locked public data of the calibration record: the NCAM 2016 assessment input series (DFO 2016, Table A2), the xteNCAM 2025 reassessment SSB series (Regular et al. 2025, Table 17), the official landings table (Regular et al. 2025, Table 1), and the capelin acoustic biomass series (observed years only). The companion forecast-ladder paper certifies the same stock's forecasting layer and its 2022–2023 observation suspension as a review-timing event; this paper certifies the viability layer. The two are the applied pair of the programme's flagship board.

0.2 Data and provenance

All inputs are the locked files of the calibration record, verified against their primary sources at lock time: `ncam_2016_table_a2.csv` (spawning stock biomass, kt, 1983–2015, cross-checked 33/33 against the primary PDF), `xtencam_table17_ssb.csv` (SSB with 95% bands and SSB-over- B_{lim} ratios, 1954–2024; implied $B_{lim} = 276$ kt), `dfo_2025_table1_landings.csv` (official removals, 1954–2023), and `capelin_acoustic_observed.csv` (observed years only, no interpolation). Two discipline rules of the lock are respected throughout: the M and F table columns are diagnostics only (dynamics are never built on them), and every annual multiplier certified below is a *net* realized quantity — it conflates the mortality regime with all removals, and is certified as such, not decomposed.

0.3 The certified windows

Proposition 1 (the two windows are sub-unitary at worst case). *From the Table A2 series: the 1983–1989 window's six annual net multipliers have minimum $\rho_g = 836.00/902.94 = \frac{41800}{45147} \approx 0.9259$ (1986), and the 1991–1993 window's two annual net multipliers are $\frac{38195}{73451} \approx 0.520$ (1992) and $\frac{2021}{7639} \approx 0.265$ (1993). Both windows' worst-case multipliers are strictly below unity.*

Proof. Each multiplier is the exact ratio of consecutive Table A2 SSB entries in kt over 100, computed in the script

as a rational; the minimum and the two bad-window values are the displayed fractions by direct comparison of the six and two ratios respectively. Each displayed fraction is below 1 because its numerator is smaller than its denominator ($41800 < 45147$, $2021 < 7639$). $\square$

Proposition 2 (the moratorium obstruction). *Across 1992–1993 the realized removals fell from 40,956 t to 11,392 t (the official landings table), approaching the closed fishery; the realized net multiplier of the same pair of years is $\frac{2021}{7639} \approx 0.265$. At the realized rates, removing the fishery did not arrest the decline: the obstruction is witnessed in the dynamics, not in the catch level.*

Proof. The removals are the certified table rows for 1992 and 1993; the multiplier is Proposition 1’s 1993 ratio, whose numerator and denominator are the realized 1993 and 1992 SSB values. A multiplier of $\frac{2021}{7639}$ with removals at 11,392 t against a biomass of tens of thousands of kt certifies that the decline is dominated by the non-fishing component of the net rate; the statement is a witness about the realized window, not a counterfactual claim about other catch paths. $\square$

0.4 Breach-time certificates

Proposition 3 (breach bounds and the realized timeline). *With $F_{\text{lim}} = 276$ kt: (i) from the end-1991 reading 734.51 kt, one step at the window’s worst multiplier already certifies breach, since $\frac{2021}{7639} \leq \frac{27600}{73451}$; (ii) under the realized two-step sequence the stock is below F_{lim} in exactly two years — $101.05 < 276$ in 1993, the xteNCAM file’s SSB-over- B_{lim} ratio 0.29 at 1993 confirming the same fact in the independent assessment; (iii) under the LRP = 884.6 kt the 1991 reading 734.51 is already below the reference point. The certificates match the realized timeline exactly.*

Proof. (i) is the exact inequality $\frac{2021}{7639} = \frac{2021 \cdot 73451}{7639 \cdot 73451}$ against $\frac{27600}{73451}$, checked by cross-multiplication of integers in the script $(2021 \cdot 73451 < 7639 \cdot 27600)$. (ii) is the direct comparison $101.05 < 276$ on the Table A2 row for 1993, with the xteNCAM ratio column read as the locked decimal 0.29 at 1993. (iii) is $734.51 < 884.6$. No step of the proof uses floating point. $\square$

0.5 The reference decade was already marginal

Proposition 4 (the LRP is the decade’s own mean, and the decade was not robustly viable). *(i) The limit reference point 884.6 kt equals the 1983–1989 mean of the Table A2 SSB series (884.58, rounded), and the decade’s readings stay within $\pm 6.4\%$ of it: the ratios run from $\frac{4180}{4423} \approx 0.945$ (1986) to $\frac{94075}{88460} \approx 1.064$ (1987). (ii) Under the window’s worst-case multiplier $\rho_g = \frac{41800}{45147}$ sustained from the 1989 reading 921.66 kt, the stock breaches $F_{\text{lim}} = 276$ kt within sixteen years: $\rho_g^{16} \leq \frac{27600}{92166} < \rho_g^{15}$. No removals policy certifies indefinite above-floor viability at worst-case realized multipliers, since even zero removals inherits the net multiplier’s strict decline.*

Proof. (i) The mean is the exact rational sum of the seven readings over seven, equal to $\frac{88458}{100}$ up to the certified rounding of the record; the displayed ratio bounds are the exact minimum and maximum of the seven ratios. (ii) The

sixteenth and fifteenth integer powers of $\frac{41800}{45147}$ are compared against $\frac{27600}{92166}$ by exact cross-multiplication in the script. The final sentence: any policy with non-negative removals has a net multiplier at most the zero-removals multiplier, which at worst case is $\rho_g < 1$, and a strict contraction crosses any fixed positive floor in finite time (at most $\lceil \log(F_{\text{lim}}/z_0)/\log \rho_g \rceil$ steps, the integer certified by the exact power comparison). $\square$

Corollary 5 (what the reference point encodes). *The LRP’s construction — the 1980s mean — is precisely the certification that the healthy decade sat at its own failure boundary: the stock was marginal at B_{lim} -scale viability before the regime shift, which is the viability reading of the same fact the assessment literature records as the LRP’s provenance.*

Proof. Immediate from Proposition 4(i)–(ii): the mean-rule fixes the boundary inside the decade’s realized range, and the worst-case contraction certifies non-viability of the sustained worst case within sixteen years. $\square$

0.6 The probe datum: monitoring, not catch

Proposition 6 (the capelin alarm leads the mortality crossing). *In the observed capelin series, the acoustic estimate falls from 5783 kt (1990) to 138 kt (1991) — a certified factor of $\frac{138}{5783} \approx 0.024$ — one step before the Table A2 mean natural mortality diagnostic crosses $M = 1$ (0.403 in 1990 to 1.002 in 1991, diagnostic column, not a model input). The one-year lead makes the covariate the certified probe datum of the regime shift: the monitoring instrument moved before the mortality regime did.*

Proof. Both readings are observed rows of the locked capelin file (no interpolation, per the lock’s discipline); the factor is their exact ratio. The mortality crossing is the diagnostic column’s consecutive values straddling 1. The proposition asserts only the temporal order of the two certified events, which the rows display directly. $\square$

Remark 7 (the monitoring consequence). Read together with Propositions 2 and 4: catch level is not the certifiable instrument at the realized rates, while the covariate alarm leads the regime crossing by one step — so the viable instrument is the monitoring schedule, exactly the programme’s review-timing reading of the 2022–2023 observation suspension certified in the companion forecast-ladder paper (§3.10 there), and the design DFO adopted in 2016 (a three-year review cycle with an interim survey indicator, with the indicator value available by early January). The programme’s monitoring-companion identities are the formal counterpart; this paper supplies their realized instance.

0.7 Delimitations

Every claim is a certificate about the realized windows of the certified series — not a forecast, not a stochastic model, and not a counterfactual catch analysis: the net multipliers conflate the mortality regime with all removals (the lock’s diagnostics rule), so no statement here identifies what a different catch path would have done. The series are assessment outputs (model-based), treated as

the locked data of record with their published uncertainty untouched. The capelin series is observed-years only; the one-year lead is a statement about the certified rows, robust to no claim about unobserved years. Single stock; two windows; no averaging beyond the LRP’s own construction.

0.8 Verification methods

Every number, fraction, inequality, and temporal claim in this paper is regenerated by the edition’s script in exact integer and rational arithmetic, standard library only, directly from the locked CSV files: the six good-window and two bad-window multipliers; the removals rows; the breach comparisons (i)–(iii) by integer cross-multiplication; the sixteenth and fifteenth powers of ρ_g ; the seven LRP ratios with their exact minimum and maximum and the mean rule; the capelin factor; and the mortality crossing. The script also asserts every headline number of this text present verbatim.

0.9 Conclusion

The obstruction programme’s machinery, applied to the locked public record of the Northern cod, certifies what the narrative record suggests and the data now proves exactly: the healthy decade was already marginal at the auxiliary threshold (the LRP is its own mean); the collapse proceeded with the fishery closing (the moratorium obstruction); the breach timeline is exactly two years from the last pre-collapse reading; and the only instrument that moved before the regime did was the monitored covariate. Viability on this stock was never a catch-optimization problem; it was a monitoring problem. That is the applied content of the obstruction programme, now certified.

References

- DFO (2016). Stock assessment of Northern cod (NAFO 2J3KL). Can. Sci. Advis. Sec. Sci. Advis. Rep. 2016/026.
- Regular, P. et al. (2025). Northern cod (NAFO 2J3KL) reassessment. DFO Can. Sci. Advis. Sec. Res. Doc. 2025/048.
- Abaee, A. The Northern cod forecast ladder under a frozen specification (the applied forecast-ladder paper, same repository), §3.10.
- Abaee, A. An obstruction calculus for viability under incomplete observation (submitted).
- Abaee, A. Probabilistic sufficiency for the obstruction calculus: belief-state safety values under partial observation (companion theory paper, same repository).

Declarations

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None.

Competing interests

None.

Data availability

All inputs are the provenance-locked public files of the repository’s calibration record (paperE1_calibration_data_v1_wave_e_cod/ and

the record’s CSV extracts), with primary sources cited at lock time.

Code availability

The verification script `applied_regime_viability_v1_verification.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it regenerates every fraction, inequality, power, ratio, and temporal claim verbatim from the locked files.

AI declaration

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